PPP — nuqta-nuqta protokoli

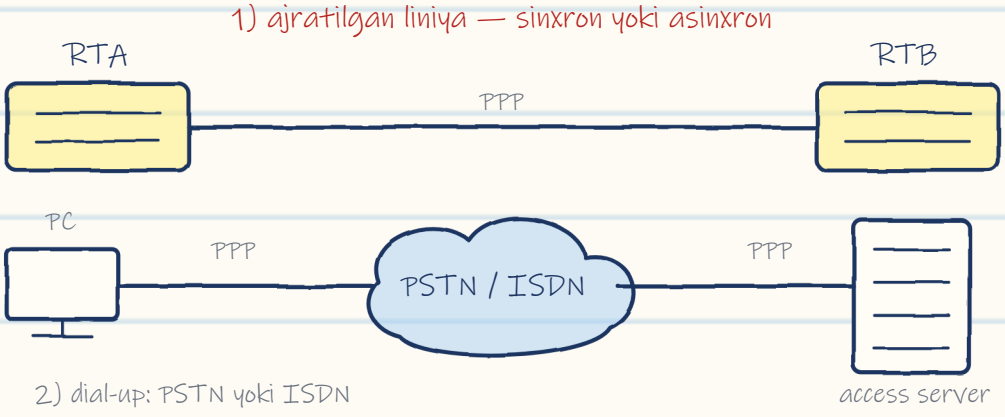
Point-to-Point Protocol

Asosiy atamalar
- PPP — nuqta-nuqta kanalda tarmoq paketlarini tashiydi
- LCP — kanalni o'rnatadi, kelishadi va uzadi
- NCP — tarmoq qatlami protokolini sozlaydi — IPCP
- PAP — parolni ochiq matnda yuboradigan autentifikatsiya
- CHAP — challenge + hash — parol liniyada ketmaydi

1. PPP nima?

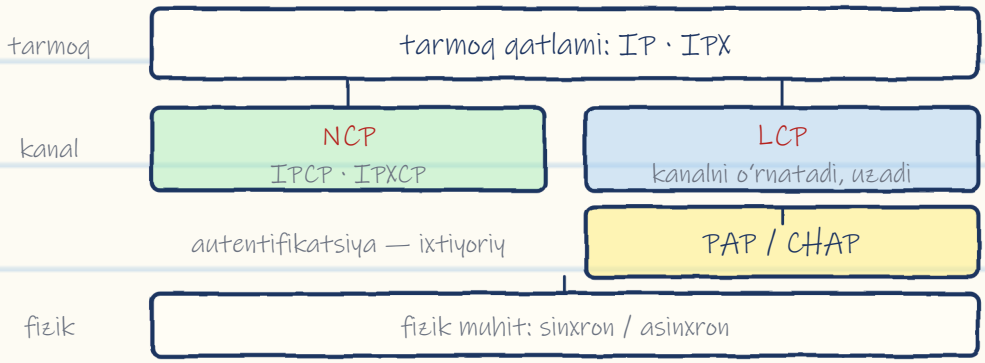
Nuqta-nuqta kanalda tarmoq qatlami paketlarini tashiydi (SLIP o'rniga)

- sinxron + asinxron — ajratilgan liniya, PSTN/ISDN dial-up
- full duplex, OSI da — kanal qatlami
- SLIP faqat asinxron edi va kelishuvni bilmasdi

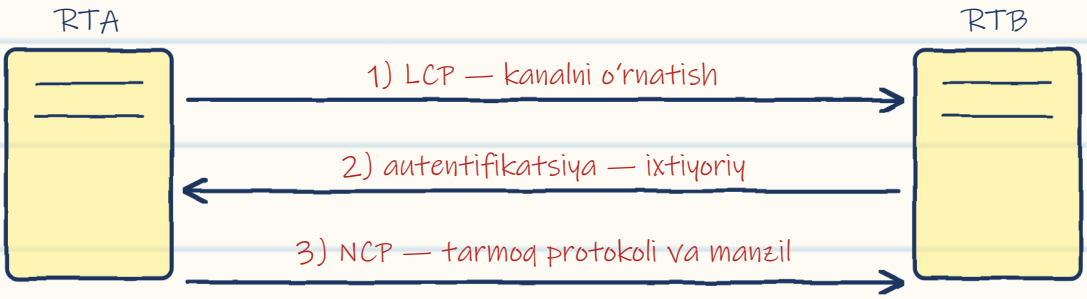


1-rasm. PPP har xil WAN kanallarida ishlaydi

2. Tarkibi



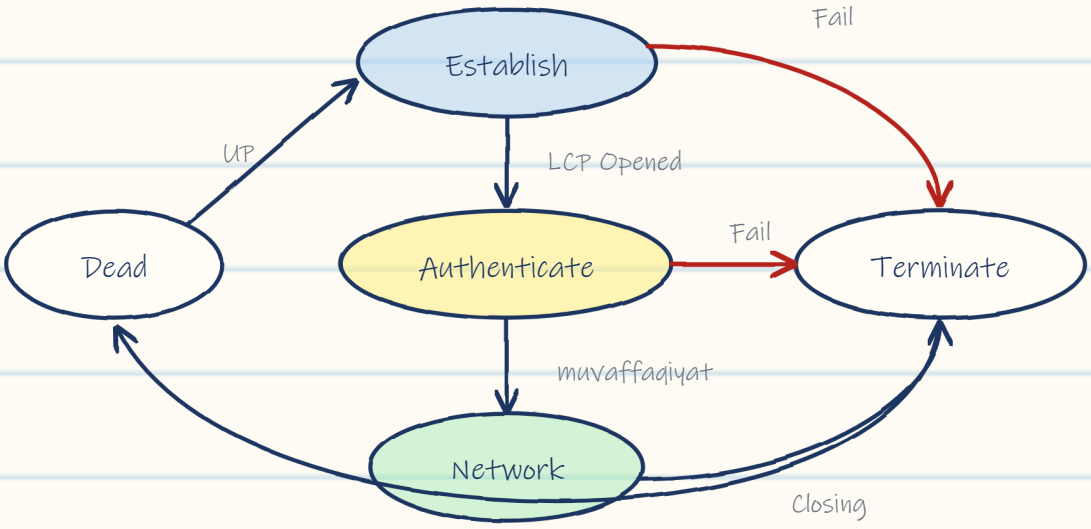
3. Sessiya bosqichlari



2-rasm. Uchala bosqich ketma-ket bajariladi

- LCP — kanal bor-yo'qligi, MTU, autentifikatsiya usuli
- autentifikatsiya — PAP yoki CHAP, shart emas
- NCP — IPCP orqali IP manzillar aniqlanadi

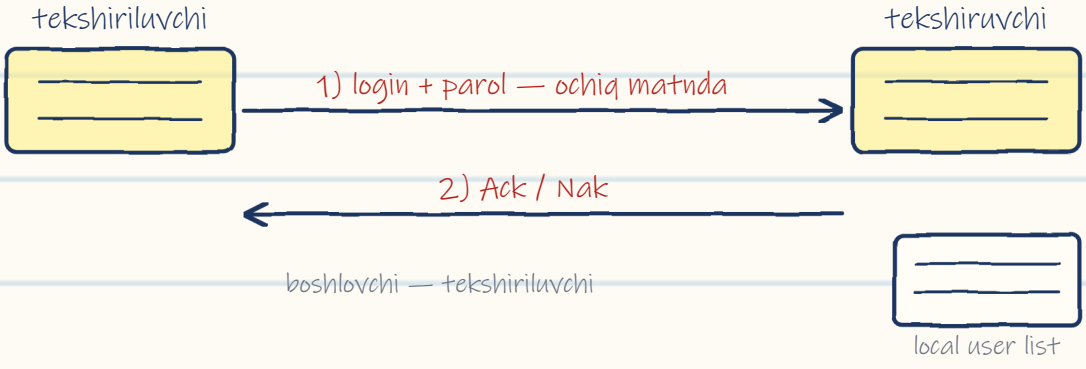
4. Holatlar diagrammasi



Down — fizik qatlam yo'q

> **Muhim:** tarmoq paketlari faqat mos NCP kelishuvidan keyin o'ta boshlaydi.

5. PAP — ikki qadam

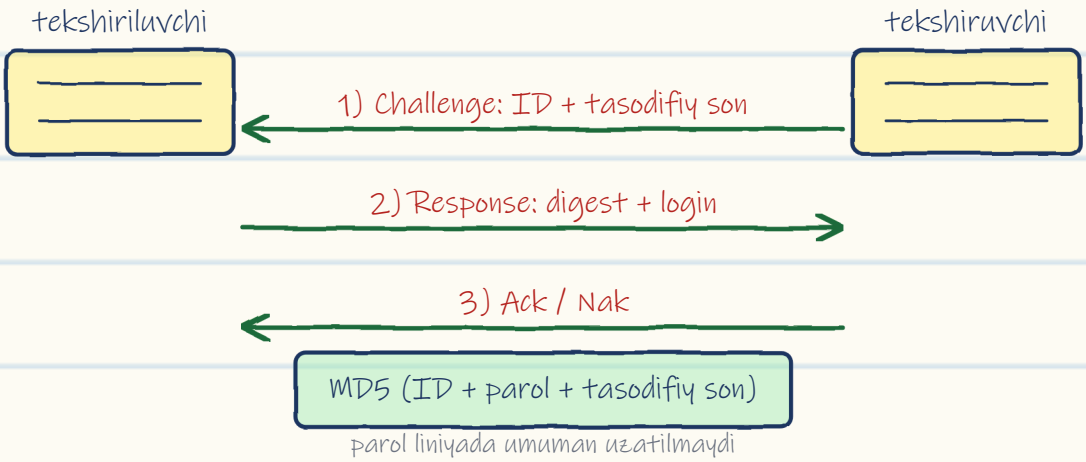


3-rasm. PAP — ikki qo'l siltash

- parol ochiq matnda ketadi — takroriy urinishdan himoya yo'q
- xato bo'lsa kanal darhol uzilmaydi — urinishlar chegarasidan keyin

Tekshiruvchi login va parolni local user list dan yoki RADIUS serveridan qaraydi.

6. CHAP — uch qadam



4-rasm. CHAP ni tekshiruvchi boshlaydi

Ikkalasi ham: bir tomonlama va ikki tomonlama ishlay oladi.

7. PAP va CHAP

	PAP	CHAP
Qadamlar	2	3
Kim boshlaydi	tekshiriluvchi	tekshiruvchi
Parol liniyada	ochiq matn	uzatilmaydi
Ikki tomonlama	bor	bor

8. PAP ni sozlash



tekshiruvchi (RTA)

```
local-user abc class network
password simple abc
service-type ppp
ppp authentication-mode pap
```

tekshiriluvchi (RTB)

```
ppp pap local-user abc
password simple abc
```

9. CHAP ni sozlash



tekshiruvchi (RTA)

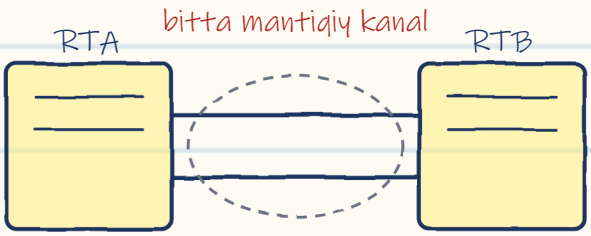
```
local-user abc class network
password simple abc
service-type ppp
ppp authentication-mode chap
ppp chap user 123
```

tekshiriluvchi (RTB)

```
local-user 123 class network
password simple 123
service-type ppp
ppp chap user abc
```

Ikkinchi yo'l: tekshiriluvchida `ppp chap password` — interfeysning standart CHAP paroli.

10. MP — Multilink PPP



Serial 2/0 + Serial 2/1

polosa · zaxira · yuk taqsimoti

fragmentatsiya — kechikish kamayadi

1) Virtual-Template

```
interface virtual-template 1
ppp mp virtual-template 1
```

2) MP-group

```
interface mp-group 1
ppp mp mp-group 1
```

5-rasm. VT — login bo'yicha bir nechta bog'lam; MP-group — bitta bog'lam

Ko'p hollarda MP-group tanlanadi — soddaroq; bog'lamga faqat fizik interfeyslar kiradi.

Xulosa

- PPP — nuqta-nuqta, sinxron va asinxron, full duplex
- Tarkibi: LCP · NCP · PAP/CHAP
- Bosqichlar: LCP → autentifikatsiya → NCP
- PAP — 2 qadam, parol ochiq matnda
- CHAP — 3 qadam, MD5, parol uzatilmaydi
- MP — bir nechta kanal bitta mantiqiy kanalda

Tekshirish

1. CHAP necha qadam? — 3
2. IP manzilni qaysi NCP kelishadi? — IPCP
3. PPP ni yoqish? — link-protocol ppp
4. MP ning 2 usuli? — virtual-template, MP-group